# THE VALUATIVE CRITERION FOR ALGEBRAIC SPACES IS PRESERVED BY BASE CHANGE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof carries the words 'This proves the lemma in the case of uniqueness' as its own transition; the existence part is then written out.

*Editorial note.* The counted spans carry no references at all, so the card needs no navigation stubs beyond the definition pointer implied by the statement shape.

*Editorial note.* Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

*Allowed standard prerequisites (editorial).*

- The shape of the valuative criterion diagram and its two parts. Definition definition-valuative-criterion, used at proof lines 8701-8711. Fixes the solid diagrams whose dotted arrows are compared.
- Universal property of the fibre product Z x_Y X. used at proof lines 8704-8710 and 8717-8725. Gives the 1-1 correspondence between the two kinds of dotted arrows.
- Valuation rings and field extensions used in the criterion. used at proof lines 8713-8715. Supplies the data produced by the existence part for f.

**Lemma 0.2.** *The base change of a morphism of algebraic spaces which satisfies the existence part of (resp. uniqueness part of) the valuative criterion by any morphism of algebraic spaces satisfies the existence part of (resp. uniqueness part of) the valuative criterion.*

*Proof.* Let $f : X \to Y$ be a morphism of algebraic spaces over the scheme $S$. Let $Z \to Y$ be any morphism of algebraic spaces over $S$. Consider a solid commutative diagram of the following shape

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \longrightarrow & Z \times_Y X & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow & \nearrow & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & Z & \longrightarrow & Y \end{array}$$

Then the set of north-west dotted arrows making the diagram commute is in 1-1 correspondence with the set of west-north-west dotted arrows making the diagram commute. This proves the lemma in the case of "uniqueness". For the existence part, assume $f$ satisfies the existence part of the valuative criterion. If we are given a solid commutative diagram as above, then by assumption there exists an extension $K'/K$

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of fields and a valuation ring $A' \subset K'$ dominating $A$ and a morphism $\mathrm{Spec}(A') \to X$ fitting into the following commutative diagram

$$\begin{array}{ccccccc} \mathrm{Spec}(K') & \longrightarrow & \mathrm{Spec}(K) & \longrightarrow & Z \times_Y X & \longrightarrow & X \\ \downarrow & & & \nearrow & & & \downarrow \\ \mathrm{Spec}(A') & \longrightarrow & \mathrm{Spec}(A) & \longrightarrow & Z & \longrightarrow & Y \end{array}$$

And by the remarks above the skew arrow corresponds to an arrow $\mathrm{Spec}(A') \to Z \times_Y X$ as desired. $\square$